

Exp No:18 Experiment to understand KNN algorithm for a given data set

Description: Understand the KNN algorithm for the dataset given.

```
In [1]: import numpy as np
import pandas as pd
```

```
In [2]: df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   sepal.length    150 non-null    float64
 1   sepal.width     150 non-null    float64
 2   petal.length    150 non-null    float64
 3   petal.width     150 non-null    float64
 4   variety         150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
In [3]: df.variety.value_counts()
```

```
Out[3]: Setosa      50
Versicolor  50
Virginica    50
Name: variety, dtype: int64
```

```
In [4]: df.head()
```

```
Out[4]:
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
In [5]: features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values
```

```
In [6]: from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

```
In [8]: xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,random_state=0)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

```
Out[8]: KNeighborsClassifier()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
In [9]: print(model_KNN.score(xtrain,ytrain))
        print(model_KNN.score(xtest,ytest))
```

```
0.9583333333333334
1.0
```

```
In [10]: from sklearn.metrics import confusion_matrix
         confusion_matrix(label,model_KNN.predict(features))
```

```
Out[10]: array([[50,  0,  0],
                [ 0, 47,  3],
                [ 0,  2, 48]], dtype=int64)
```

```
In [11]: from sklearn.metrics import classification_report
         print(classification_report(label,model_KNN.predict(features)))
```

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.94	0.95	50
Virginica	0.94	0.96	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

```
In [ ]:
```